Health Monitoring System - Costing

Sr.No.	Components	Cost
1.	ESP32 Development Board	350
2.	Pulse Sensor	200
3.	DS18B20 Temp. Sensor	180
4.	16*2 LCD Display	90
5.	I2C Module For LCD	80
6.	Power Supply / USB Cable	100
7.	Miscellaneous Components	200
	TOTAL	**1200**

CONCLUSION:

- In this project, we successfully built a simple and affordable health monitoring system using the ESP32 microcontroller. The system is able to measure heart rate and body temperature and display the values on an LCD screen as well as on a web page through WiFi.
- the system proves that with low-cost components and basic programming, we can create a useful health monitoring solution that can be helpful for home use and basic patient monitoring. This project improved our practical knowledge in embedded systems and IoT applications.

FUTURE SCOPE:

1. Additional health sensors such as a blood oxygen (SpO_2) sensor, ECG sensor, or blood pressure sensor can be added.
2. The system can also be integrated with a mobile application to provide a better user interface and real-time notifications. Emergency alerts such as SMS or email can be added if the heart rate or temperature crosses safe limits.
3. artificial intelligence and data analysis techniques can be used to detect abnormal patterns and predict possible health issues early.